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CO₂ Emissions Reduction in Onshore Mahakam Drilling Through the First Hybrid PDC Bit Application: Eliminating Multiple Bit Runs and Increasing ROP in Hard Interbedded Formations

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Abstract

Idle since 2014 due to marginal economics and complex drilling conditions, Badak North (BDK-N) was revisited in 2025 with two cluster wells (N-1 and N-2) targeting cost-effective, low-emission delivery. The intermediate section presented interbedded formations with 20,000 psi stringers, long intervals, and repeated bit failures—driving inefficiency and high CO₂ output. This paper details the first successful Hybrid PDC bit deployment in the field, completing the longest 12-1/4" interval in a single run, boosting drilling efficiency and reducing emissions by 36%.

The N-1 well encountered challenges—prolonged 16.4 drilling days, slow penetration rate (29.1 ft/hr), four unplanned bit trips, and excessive CO₂ emissions—due to hard interbedded sandstone formations with UCS reaching 20,000 psi. Based on offset well drilling parameter analysis, rock strength modeling, and bit reports, a hybrid PDC bit that integrates roller cone shearing and PDC crushing mechanisms was theoretically suitable for the N-2 well. A detailed bit roadmap serves as a drilling parameter recommendation and keeps bearing integrity. Real-time drilling parameters and vibration monitoring were executed to optimize bit performance and mitigate bearing failure while enhancing steering control in complex lithologies.

The hybrid PDC bit achieved a 68.1 ft/hr ROP—2.5 times faster than the previous average—eliminating four trips and saving eight rig days. It drilled 6,911 ft in a single run, marking the longest and deepest 12-1/4" section in Badak Field history. The run also became the fastest intermediate section drilled in the 2024-2025 campaign. Notably, the bit remained in-gauge and green-graded despite surpassing its bit revolution limit by 28% without any bearing failure. Following verification against internal company protocols and international emissions standards, the N-2 well achieved a CO₂ emission reduction of 126 metric tons—representing a 36% decrease compared to the N-1 well. Operational cost savings reached US \$600,000 while maintaining safety aspects.

This paper introduces a field-proven methodology for achieving lower-emission drilling in hard interbedded formations using a single bit run. It offers a replicable example of how formation analysis, pre-job planning, and execution monitoring can collectively contribute to carbon reduction, improving operational efficiency and economic benefits in marginal fields.

Introduction

The Badak Field in East Kalimantan, operated by Pertamina Hulu Sanga Sanga (PHSS) since 2018, is one of the world's largest gas fields. Discovered in 1972 and onstream since 1976, it peaked at 1,100 MMSCFD in 1990 and has produced over 6 TCF of gas to date (Wibowo et al., 2012). Now a mature asset producing 15.6 MMSCFD from 274 wells, it requires continuous drilling and interventions due to unlock its thin, interbedded sandstone reservoirs.

The Badak North (BDK-N) area, two idle wells drilled in 2014, was redrilled in late 2024 with BDK N-1 and N-2 targeting undeveloped reservoirs under depleted conditions. BDK N-1 was safely delivered but faced prolonged drilling time and high CO₂ emissions from repeated bit failures in the 12-1/4" intermediate section, which contained interbedded formations and hard stringers with UCS up to 40,000 psi.

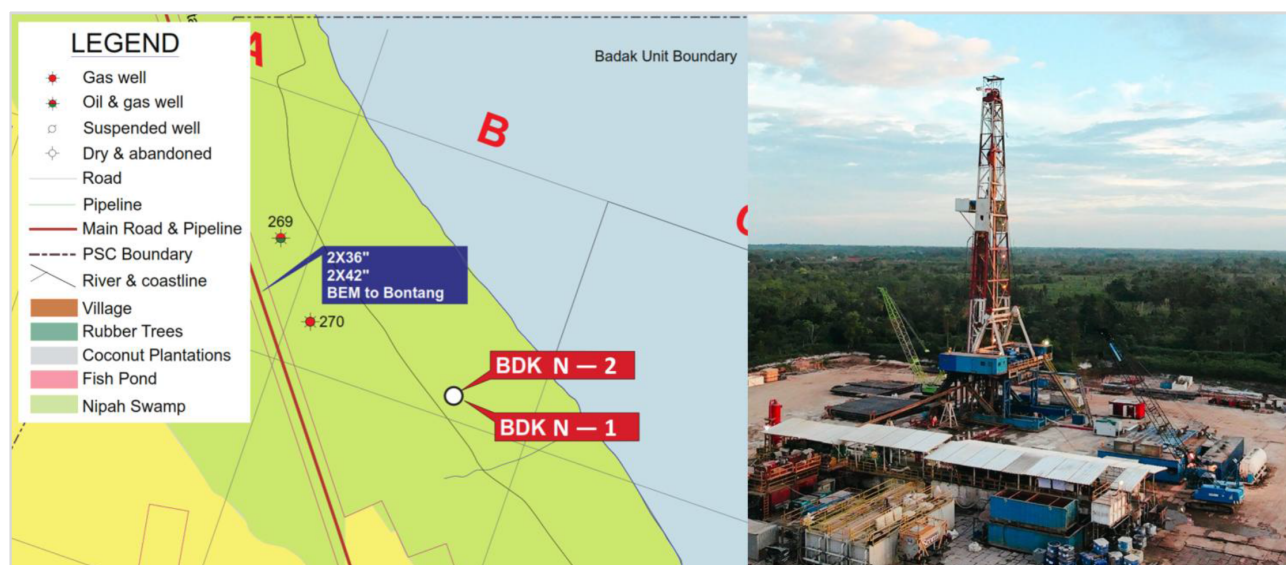


Figure 1—Location of the Badak North area showing the BDK N-1 and BDK N-2 cluster wells, with a 2,000 HP drilling rig in operation.

These lessons drove a specific strategy for BDK N-2, completing the first successful Hybrid PDC bit application in the Badak Field. The approach achieved the deepest 12-1/4" intermediate hole in field history, reduced drilling emissions by 36%, and improved operational efficiency—demonstrating how bit technology, formation analysis, and execution monitoring can align technical performance with environmental goals.

Lessons from the First Well (BDK N-1)

Operation summary

The BDK N-1 well targeted maximum gas deliverability in the scarce Badak North area, with a planned total depth of 13,052 ftMD in three hole sections (Fig. 2). The most challenging was the 12-1/4" intermediate section, 6,564 ftMD long, containing two of the eight reservoir layers and interbedded formations with UCS ranging from 6,000 to 40,000 psi.

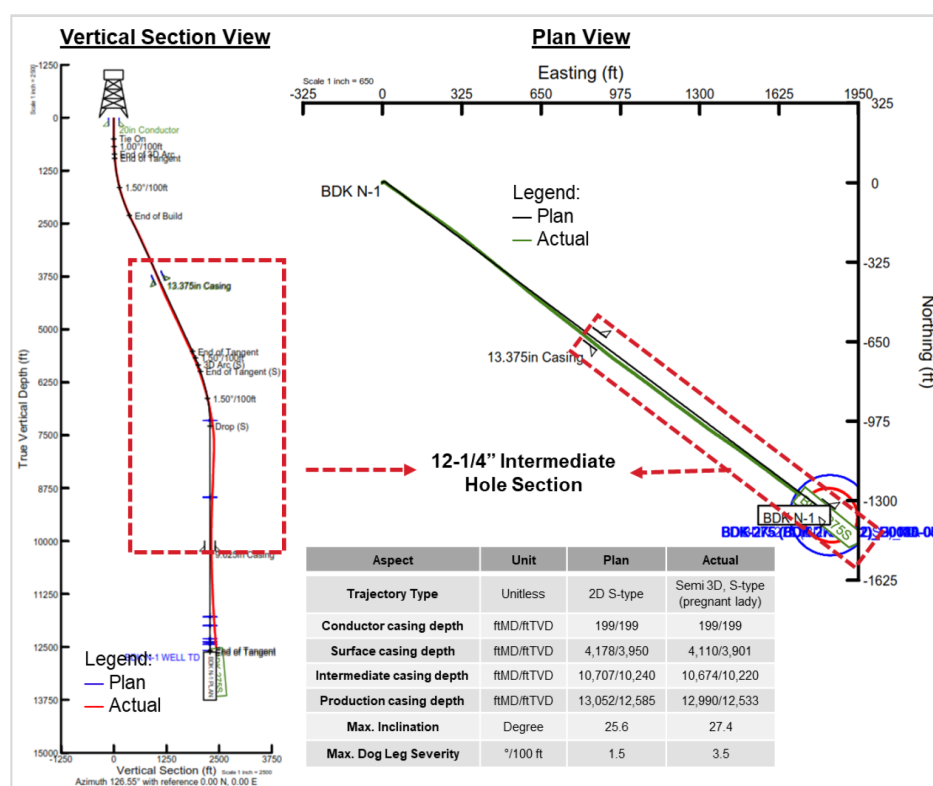


Figure 2—Planned and actual trajectories of the BDK N-1 well, highlighting the 12-1/4" intermediate hole section. The deviation from the plan resulted a "pregnant lady" shape between 7,500 and 10,000 ftTVD, slowed drilling progress and increased carbon emissions.

This interval took 16.5 days to drill, 7.3 days longer than planned, due to multiple unplanned bit trips and minor rig equipment issues. Average trip time was 1.5 days, intensifying delays and CO₂ emissions. Mismatch between bit design, formation properties, and directional tools limited ROP, reducing operational efficiency and profitability. The section generated 351.2 tCO₂e emissions.

Table 1—BDK N-1 General Well Data Showing Planned versus Actual Parameters and Drilling Duration Gap.

Aspect	Unit	Plan	Actual
Architecture	Unitless	Three sections	
Hole size & casing size	Inch	17-1/2" hole & 13-3/8" casing, 12-1/4" hole & 9-5/8" casing, 8-1/2" hole & dual 3-1/2" tubing	
Completion type	Unitless	Dual 3-1/2" monobore tubing	
Trajectory type	Unitless	S-type two dimension (2D)	
Well total depth	ftMD/ftTVD	13,052/12,585	12,990/12,533
Intermediate hole length	ftMD	6,529	6,564
Total drilling duration	days	46.8	63.1
Intermediate hole section, drilling duration	days	9.2	16.5
Intermediate hole section, drilling duration gap	days	N/A	7.3

Progression charts (time–depth plots) were utilized to compare drilling performance benchmarks (Rushmore et al., 2011). The planned 12-1/4" section trajectory deviated significantly, with actual progress being on average 308 ft/day slower than planned. The section took 26.1% of total drilling duration compared to the planned 19.6%, driven by three unplanned trips due to low ROP and steering inefficiency.

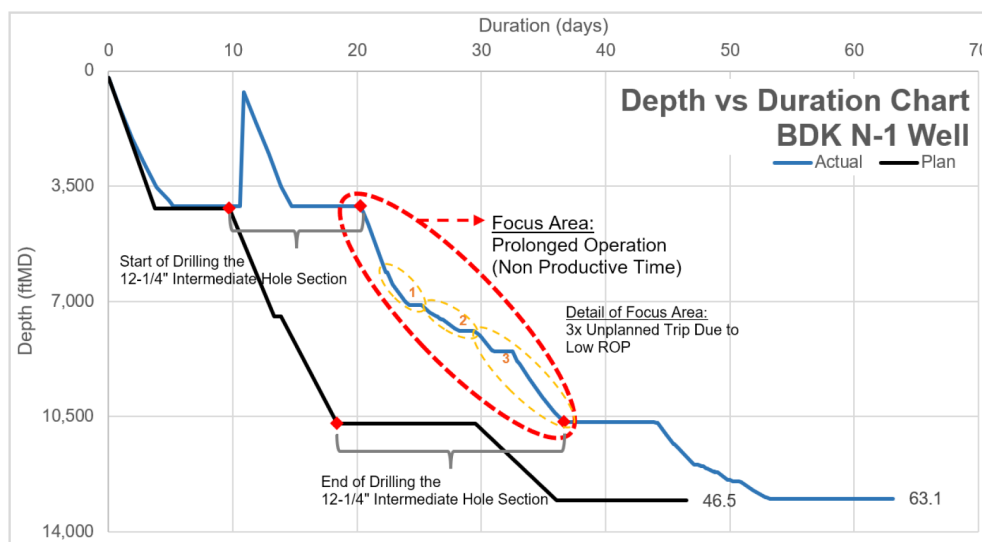


Figure 3—Depth vs. Duration (progression) chart for the BDK N-1 well, showing a significant deviation between planned and actual lines in the 12-1/4" intermediate hole section. The gap was primarily caused by low ROP and three unplanned trips.

Operational anomalies appeared at ~6,000 ftMD, where ROP decreased from ~100 to <60 ft/h. A temporary recovery to 100 ft/h was achieved by increasing WOB to 12 klbs, but ROP quickly dropped below 30 ft/h as DSE spiked, indicating inefficient drilling. The first unplanned trip occurred at 7,112 ftMD; subsequent trips at 7,886 ftMD and 8,521 ftMD were triggered by persistently low ROP during high sliding percentages (78.0% and 42.4%).

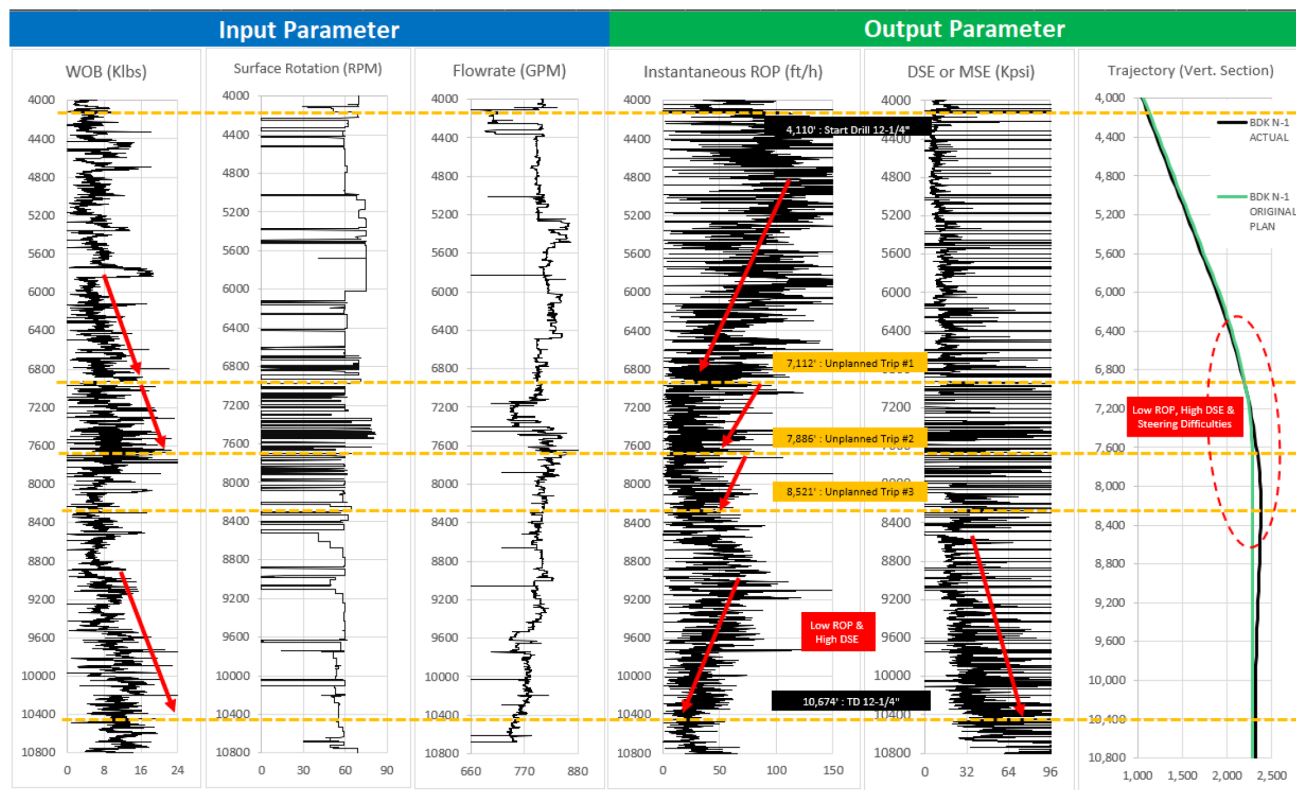


Figure 4—Depth-based plot of actual drilling parameters, showing controllable inputs (WOB, surface RPM, and flow rate) and corresponding output (instantaneous ROP, DSE, and trajectory).

Drilling Specific Energy (DSE), which consists axial, torsional, and hydraulic energy, was used to assess efficiency. Unlike MSE, DSE accounts for hydraulics via Teale's (1964) formula modified by Armenta (2008). Higher DSE values in drilling intervals correlated with ROP reduction that triggered the unplanned trips, and can be expressed as:

$$DSE = \frac{WOB}{A_B} + \frac{120 \times \pi \times RPM \times T}{A_B \times ROP} - \frac{1,980,000 \times \lambda \times HP_B}{ROP \times A_B}$$

The first two terms are consistent with Teale's original equation, while the third term accounts for bit hydraulics. The factor 1,980,000 is a unit conversion constant, and λ is a dimensionless hydraulic factor (0.09 for a 12-1/4" bit). The ratio of bit hydraulic power to bit area (HP_B/A_B) defines the bit hydraulic horsepower per square inch (HSI), while ROP is the rate of penetration (ft/h).

Post-trip analysis concluded bit dull condition and enormous trajectory corrections as the main causes. With optimized parameters (WOB 18–24 klbs, variable RPM/flow rate), sliding mode limited ROP to 20–60 ft/h. The final run, using a rerun bit with minimal sliding, improved ROP to 50–100 ft/h and increased the progression gradient, though the last 2,000 ft showed rising DSE and falling ROP in harder formations.

Drilling Bit Run Summary

Table 2—BDK N-1 drilling bit run summary

Unplanned trip no.	1	2	3	4 (planned due to reaching TD)
Steering type	Mud Motor	Mud Motor	Mud Motor	Mud Motor
Bit type	PDC (New)	PDC (New)	PDC (Rerun from No. 2)	PDC (Rerun from No. 3)
IADC classification	M422	M423	M423	M423
Number of blades	6	7	7	7
Cutter size (mm)	16	16	16	16
Total flow area (in ²)	1.5102	1.5309	1.5309	1.5309
Depth in (ftMD)	4,116	7,112	7,886	8,521
Depth out (ftMD)	7,112	7,886	8,521	10,674
Drilled footage (ftMD)	2,996	774	635	2,153
Feet rotary (ftMD)	2,511	170	366	2,043
Feet steered (ftMD)	485	604	269	110
Steering to total drilled footage (%)	16.2%	78.0%	42.4%	5.1%
Average on bottom ROP (ft/h)	42.6	23.2	22.4	19.2
Bit dull grading in	N/A (new)	N/A (new)	1-1-WT-G-X-1/16-NO-ROP	1-2-WT-G-X-1/16-NO-ROP
Bit dull grading out	1-2-WT-G-X-2/16-CT-ROP	1-1-WT-G-X-1/16-NO-ROP	1-2-WT-G-X-1/16-NO-ROP	2-2-CT-G-X-3/16-WT-TD

Two PDC bit types were used in the 12-1/4" open-hole section: M422 (six blades, short profile) and M423 (seven blades, medium profile). Higher blade counts typically improve durability but reduce aggressiveness;

fewer blades drill faster but less durable. Per IADC classification, both are matrix-body bits for hard formations with 14–24 mm cutters.

The first run with M422 drilled the longest and fastest in single run, drilling mostly softer formations (5,000–10,000 psi UCS) and encountering only a few medium-strength stringers (~25,000 psi UCS) at 6,450 and 6,700 ftMD. Dull grading showed cutters in good condition but a 2/16" under-gauge body, likely from abrasive wear, which reduced penetration rate. ROP significantly decreased from ~100 ft/h to 30 ft/h despite parameter adjustments, triggering the urgency of trip.

M423 bits were utilized in the second and third runs for remaining footage and trajectory corrections, with sliding comprising 78.0% and 42.4% of footage, respectively. This severely slow ROP, which declined from 100 ft/h to 30 ft/h despite higher WOB, varied RPM, and optimized flow rate. Both runs were stopped early due to persistently low ROP, high DSE, and steering challenges.

The fourth run reused the bit from the third run, considered in a good condition for rerun. Less sliding (5%) improved efficiency, and the bit drilled 2,153 ft to TD. However, average on-bottom ROP was only 19.2 ft/h, with performance declining in the final 1,000 ft.

While the M423 demonstrated durability, its insignificant aggressiveness reduced operational efficiency, increased cost, and contributed to higher CO₂ emissions. These results drove the assessment of alternative bit designs combining durability with high ROP for extended single-run drilling in similar formations.

Formation Review

The BDK N-1 and N-2 cluster wells present lithology consistency with the Badak North Area: thin (5–15 ft) interbedded sequences dominated by sandstone, shale, coal, and marine carbonates, indicating the deltaic depositional environment of the Badak Field with scattered sand-lens reservoirs.

The UCS profile in [Figure 5](#) integrates cuttings analysis with wireline logs (gamma ray, resistivity, porosity, and density). Formations are classified as soft (<10,000 psi), medium (10,000–20,000 psi), and hard (>20,000 psi). The first unplanned trip mostly drilled soft formations, with some carbonate stringers (~25,000 psi) encountered near interval end. The most challenging zones were high-UCS carbonate stringers (highlighted in red), which reduced ROP significantly, increased bit wear, and followed the logical trend where deeper, harder formations result in slower ROP and greater wear ([Mayibeki et al., 2023](#)).

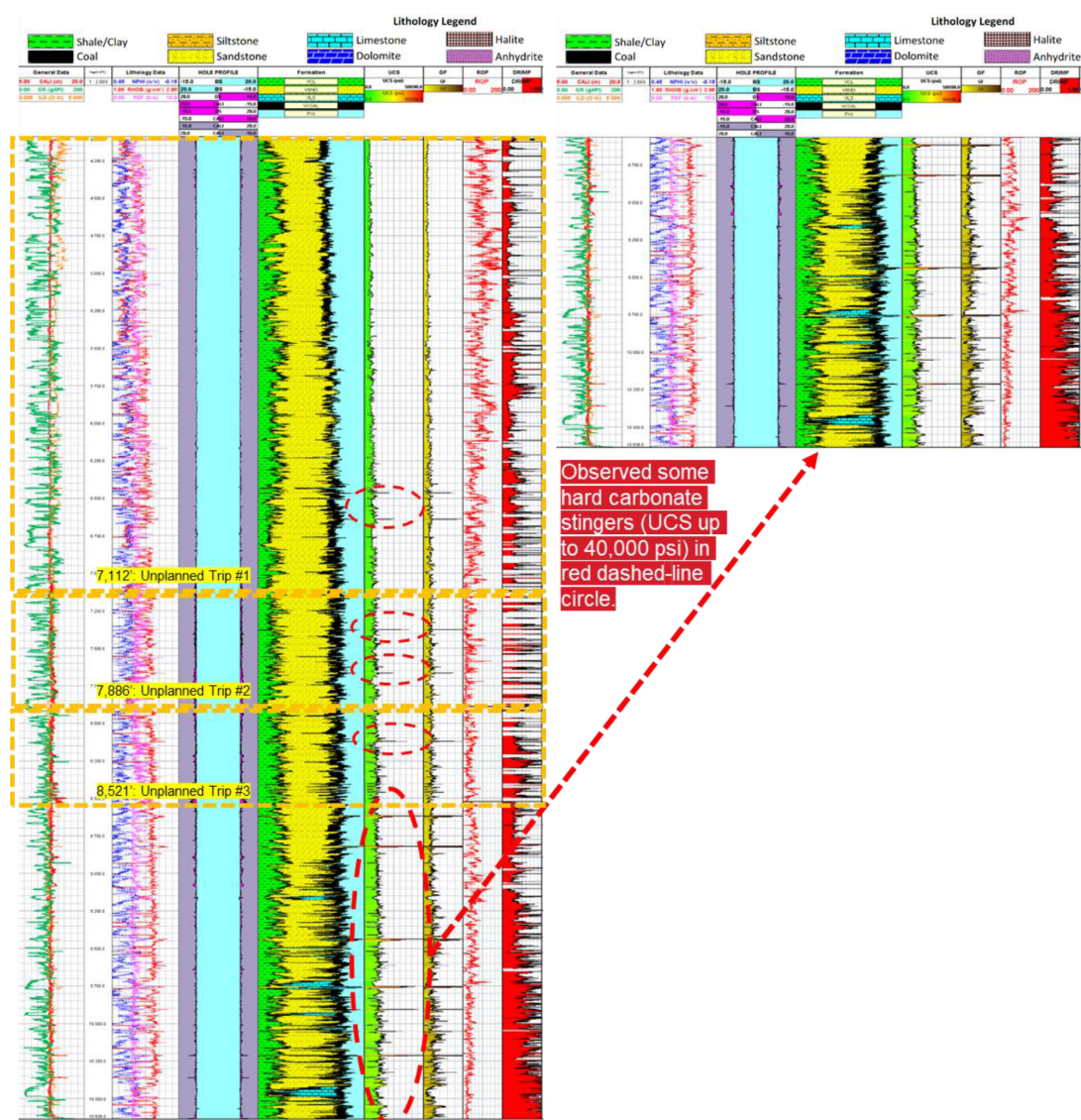


Figure 5—Badak N-1 12-1/4 in. hole section formation and rock strength (UCS) prediction from wireline logging and drilling cuttings data. UCS ranges from 10,000–20,000 psi, with carbonate stringers up to 40,000 psi

Correlation of UCS with drilling parameters and efficiency metrics (e.g., DSE) confirmed this trend, ROP dropped and DSE spiked when drilling through hard stringers, even with increased WOB. After penetrating a stringer, ROP rarely recovered to previous levels, indicating cumulative bit damage. These findings justified the need for bit designs capable of maintaining performance during drilling rapid lithological shift in the Badak North area.

The BDK N-2 wellhead was located only 60 ft from BDK N-1, registered a similar geological characteristic. The well was planned with an S-type 2D trajectory, drilling directionally northeast to converge from the actual BDK N-1 trajectory, which trended southeast as shown in Figure 6.

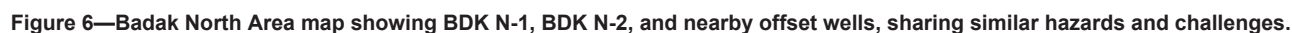


Table 3—BDK N-2 general well data compared with BDK N-1, highlighting the planned 676 ftMD deeper 12-1/4 in. intermediate casing depth than N-1.

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Hybrid PDC Bit Trial & Strategy

Theoretical Approach

Bit selection for the BDK N-1 and BDK N-2 campaign was justified by formation analysis and offset well performance review. Historical data indicated that conventional PDC bits underperformed in intervals with UCS up to 30,000 psi, often requiring multiple runs and embracing severe body and cutter damage.

A crush-and-shear hybrid PDC bit was identified as the optimal solution. This design combines the shearing action of PDC cutters with the impact resistance of rolling cone elements, allowing improved rock-breaking efficiency in rapid formations shift. Key features include advanced gauge protection, optimized nozzle placement for efficient cuttings evacuation, and reinforced blades for high-impact resistance. The hybrid design provides smoother torque response, extended bit life, and higher ROP in lithologies where conventional PDCs typically struggle.

Technical Limit

Given the target drilling interval, several improvements were applied into the original bit design to ensure durability and performance. The shoulder cutter layout was reinforced by prioritizing cutter placement in the highest impact-loading areas. This configuration not only improved distance drilled but also enhanced ring-out resistance. By optimizing the cutter arrangement, loads on the cone and shoulder areas were reduced, improving the bit's ability to endure the abrasive and interbedded formations encountered in BDK N-2.

The PDC cutting structure was optimized by adopting a three-blade-to-center design. This geometry improved core-out resistance and overall drilling efficiency, particularly in formations prone to rapid lithological changes. The design sought to maximize rock destruction per revolution while maintaining structural integrity over the long interval.

A key operational concern when applying hybrid PDC bits over extended depths is the potential to exceed the maximum revolution limitation, as moving parts introduce a higher risk of mechanical failure. To mitigate the risk of cone loss due to bearing or seal failure, an additional cone guard retention mechanism was integrated into the bit body. This system retains the cone between the bearing pin and the bit body, providing redundancy against cone loss and reducing the likelihood of resulting non-productive time (NPT).

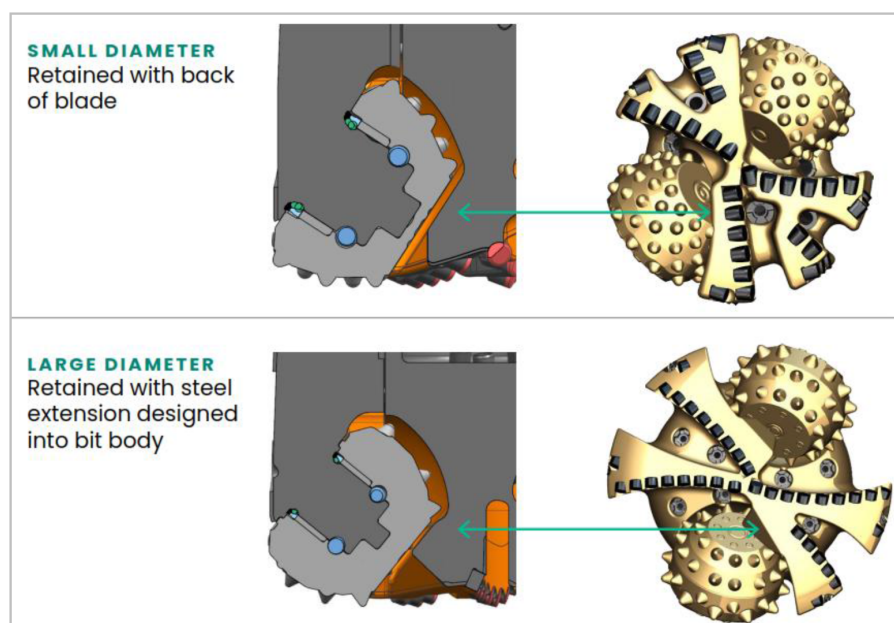


Figure 7—Hybrid PDC bit cone guard retention mechanism.

Bit Roadmap & Drilling Parameter Recommendation

The bit roadmap for BDK N-2 was developed based on the optimum operating parameters for the 12-1/4" hybrid PDC bit, with consideration for the field's lithology and past performance in the Badak North Area. The target interval is predominantly interbedded shale and sandstone, with localized coal and limestone stringers. This variability demands a parameter set that balances rate of penetration (ROP), bit life, and stability.

The recommended operating range for weight on bit (WOB) is 25–60 klbs, with adjustments made in real time based on axial vibration, ROP response, and hole conditions. Baseline flow rates are set at 650–850 GPM, paired with a motor speed of 0.15 rev/gal, resulting in a motor RPM of 90–120. With a surface RPM baseline of 90–180, the combined bit RPM is expected to range between 180–240. The maximum allowable WN value—defined as $WOB \times \text{bit RPM}$ —is 7,500, and the 80% reliability limit for the bit's cumulative revolutions (Krev) is 960.

		Bit RPM															
		100	110	120	130	140	150	160	170	180	185	190	200	210	220	230	240
WOB	10	1000	1100	1200	1300	1400	1500	1600	1700	1800	1850	1900	2000	2100	2200	2300	2400
	15	1500	1650	1800	1875	1950	2025	2100	2175	2250	2325	2400	2475	2550	2625	2700	2775
	20	2000	2200	2400	2500	2600	2700	2800	2900	3000	3100	3200	3300	3400	3500	3600	3700
	25	2500	2750	3000	3125	3250	3375	3500	3625	3750	3875	4000	4125	4250	4375	4500	4625
	30	3000	3300	3600	3750	3900	4050	4200	4350	4500	4650	4800	4950	5100	5250	5400	5550
	35	3500	3850	4200	4375	4550	4725	4900	5075	5250	5425	5600	5775	5950	6125	6300	6475
	40	4400	4600	4800	5000	5200	5400	5600	5800	6000	6200	6400	6600	6800	7000	7200	7400
	45	4950	5175	5400	5625	5850	6075	6300	6525	6750	6975	7200	7425	7650	7875	8100	8325
	50	5500	5750	6000	6250	6500	6750	7000	7250	7500	7750	8000	8250	8500	8750	9000	9250
	55	6050	6325	6600	6875	7150	7425	7700	7975	8250	8525	8800	9075	9350	9625	9900	10175
	60	6600	6900	7200	7500	7800	8100	8400	8700	9000	9300	9600	9900	10200	10500	10800	11100

Figure 8—WN quick reference table for 12-1/4" hybrid PDC bit, the maximum WN is 7,500.

During drilling, axial and lateral vibrations should be closely monitored, with axial vibration mitigation prioritized. In most cases, increasing WOB above baseline is preferred to control axial motion while maintaining drilling efficiency. When reaming a stand, RPM should be reduced to 50–60 in both upward and downward movements, avoiding contact with the bottom without rotation to minimize the risk of damaging the bit against ledges or the low side of the borehole.

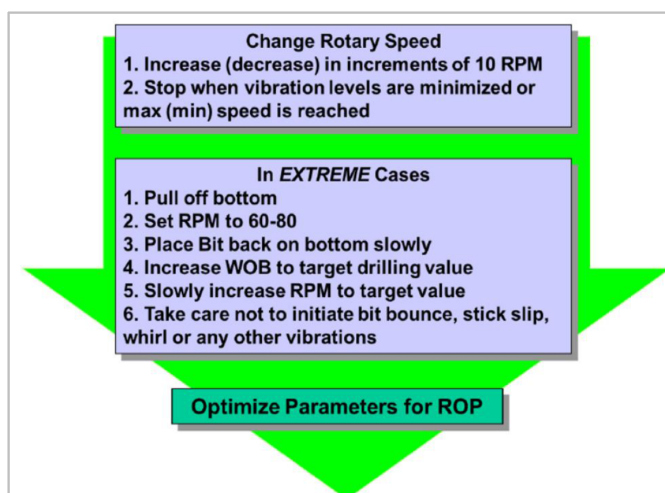


Figure 9—The recommended mitigation practices when axial vibration is observed

Mechanical Specific Energy (MSE) should be tracked continuously, with increases of three times or more above baseline considered a warning sign of reduced efficiency or potential bit damage. Exceeding the 960 Krev limit requires close monitoring for specific indicators:

- Torque or Differential Pressure: A rapid increase (1.3–2X baseline) warrants pulling out of hole (POOH).
- MSE: A sharp rise (1.5–2X) without a change in formation suggests impending bit failure and justifies POOH.
- Friction Coefficient ($\mu = Tq/WOB$): A significant increase without lithology change is also a POOH trigger.
- ROP Trends: If ROP declines in the same formation without MSE increase, conduct a drill-off test. A positive ROP response to increased WOB indicates remaining cutting structure, whereas no improvement suggests impending bit inefficiency. In such cases, revert to previous parameters, monitor for one hour, and if no recovery is observed—POOH. Notably, a short-lived ROP increase before rapid decline often precedes catastrophic bit failure.

By following this roadmap, the hybrid bit can be operated within its mechanical limits, mitigating risk while maximizing performance across the long, lithologically complex interval.

BDK N-2 Operation & Post-run Analysis

Drilling Duration Benchmark

The chart illustrates the performance difference in drilling the 12-1/4" hole section between BDK N-1 and N-2. Drilling gradient, defined as feet drilled per day, was significantly higher in N-2, averaging 813 ft/day—203% greater than N-1's 400 ft/day. The absence of unplanned trips in N-2 enabled this doubled gradient, producing a steeper performance curve compared with N-1.

Lessons from operational inefficiencies in N-1 guided bit selection and risk mitigation in N-2. While N-1 relied on conventional PDC bits, N-2 deployed a Hybrid PDC bit to address hard-formation challenges. As a result, N-2 completed the 12-1/4" section in a single bit run just only 8.5 days, 7.9 days faster than plan and 1.4 times faster than N-1, while drilling 676 ft deeper. This approach improved efficiency and reduced carbon emissions compared to the previous well.

Depth-based penetration rate analysis (Figure 10) confirms that BDK N-2's performance gains were closely linked to formation response and optimized drilling parameters. The hybrid PDC bit consistently

sustained higher instantaneous ROP, up to 2.5 times that of BDK N-1, while maintaining lower DSE values, even when penetrating high-UCS carbonate stringers. This reflects more efficient rock failure and reduced energy emittance compared to conventional PDC performance in offset wells.

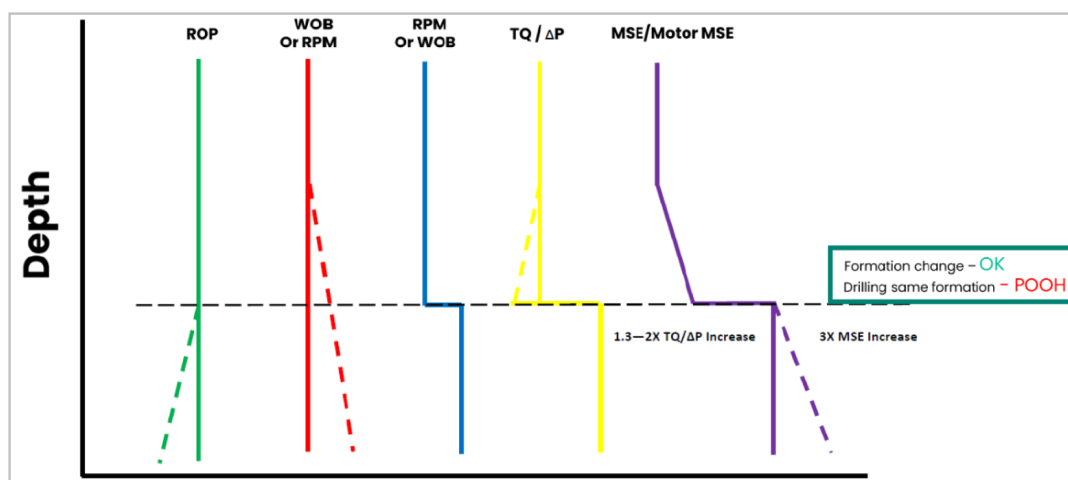


Figure 10—Trendline illustration of MSE (or DSE) as a quick-reference guide for detecting drilling anomalies and differentiating formation changes from same-formation inefficiencies.

The ability to maintain drilling efficiency across various lithologies confirmed the record single bit run and faster section progress, reducing operational time and associated CO₂ emissions.

Benchmarking to Another Wells in Badak Field

From Table 4, BDK N-2 achieved the deepest 12-1/4" hole section TD in the Badak North Area, exceeding two offset wells drilled by the previous block operator in 2014. While BDK N-2 shared similar design concepts with BDK N-1, its deeper 12-1/4" section was planned to clearly separate the hydrostatic and overpressure regimes based on pore pressure and fracture gradient (PPFG) data. Benchmarking against four other wells, including those drilled with a single bit, showed that their ROPs were significantly lower, in some cases achieving only half of BDK N-2's rate. Although BDK-270 drilled a longer interval by footage, it required two bit runs. BDK N-2 not only reached the deepest 12-1/4" section in the area but also recorded the fastest penetration rate.

Table 4—Performance metrics for drilling the 12-1/4" section in Badak North Area.
BDK N-2 achieved the greatest depth and the highest average on-bottom ROP.

12-1/4" Hole Section	Unit	BDK-269	BDK-270	BDK N-1	BDK N-2	Note
Year drilled	year	2014	2014	2025	2025	
Start depth	ftMD	3,790	3,205	4,110	4,365	End of previous section depth
End depth	ftMD	10,740	10,550	10,674	11,276	Red: deepest, green: shallowest
Footage	ftMD	6,950	7,345	6,564	6,911	End minus start depth
Number of bits run	Number	1	2	4	1	Red: more (bad), green: less (better)
Average On Bottom ROP	ft/h	34.0	41.5	41.4	68.1	Red: slowest, green: fastest
Bit type	N/A	PDC	PDC	PDC	Hybrid PDC	

Table 5—BDK N-1 & N-2 drilling bit run summary

Bit trip no.	1 st Badak N-1	2 nd Badak N-1	3 rd Badak N-1	4 th Badak N-1	1 st Badak N-2
Trip job type	Unplanned	Unplanned	Unplanned	Planned	Planned
Steering type	Mud Motor	Mud Motor	Mud Motor	Mud Motor	Mud Motor
Bit type	PDC (New)	PDC (New)	PDC (Rerun from No. 2)	PDC (Rerun from No. 3)	Hybrid (PDC & Roller Cone)
IADC classification	M422	M423	M423	M423	N/A (Hybrid PDC)
Number of blades	6	7	7	7	5 (PDC) + 2 (Cone)
Cutter size (mm)	16	16	16	16	16
Total flow area (in ²)	1.5102	1.5309	1.5309	1.5309	1.493
Depth in (ftMD)	4,116	7,112	7,886	8,521	4,365
Depth out (ftMD)	7,112	7,886	8,521	10,674	11,276
Drilled footage (ftMD)	2,996	774	635	2,153	6,911
Feet rotary (ftMD)	2,511	170	366	2,043	5,659
Feet steered (ftMD)	485	604	269	110	1,252
Steering to total drilled footage (%)	16.2%	78.0%	42.4%	5.1%	18.1%
Average on bottom ROP (fph)	42.6	23.2	22.4	19.2	68.1
Bit dull grading in	N/A (new)	N/A (new)	1-1-WT-G-X-1/16-NO-ROP	1-2-WT-G-X-1/16-NO-ROP	N/A (new)
Bit dull grading out	1-2-WT-G-X-2/16-CT-ROP	1-1-WT-G-X-1/16-NO-ROP	1-2-WT-G-X-1/16-NO-ROP	2-2-CT-G-X-3/16-WT-TD	Cone: 0-1-WT-A-E-I-NO-TD PDC: 0-1-WT-S-X-I-CT-TD

Across the 2024–2025 Badak Field drilling campaign, seven wells were drilled, five of which were directly comparable due to similar well architecture. In these comparable wells, the average 12-1/4" section length was approximately 6,000 ft. Three of them required multiple bits run, incurring higher costs and additional CO₂ emissions, while BDK 24-3 achieved a single run but only covered 2,532 ft. As shown in Table 6, all wells with inefficiencies were drilled with conventional PDC bits. In contrast, the Hybrid PDC deployed in BDK N-2 delivered a single-run completion of the deepest 12-1/4" section while achieving the fastest ROP among all wells in the campaign.

Table 6—Performance metrics for drilling the 12-1/4" section across the 2024–2025 Badak Field campaign. Hybrid PDC bit in BDK N-2 delivered the deepest depth and fastest average on-bottom ROP.

12-1/4" Hole Section	Unit	BDK 24-1	BDK 24-2	BDK 24-3	BDK N-1	BDK N-2	Note
Year drilled	year	2024	2024	2024	2025	2025	
Start depth	ftMD	4,300	3,474	4,143	4,110	4,365	End of previous section depth
End depth	ftMD	10,258	10,935	6,675	10,674	11,276	Red: deepest, green: shallowest
Footage	ftMD	5,958	7,461	2,532	6,564	6,911	End minus start depth
Number of bits run	Number	2	3	1	4	1	Red: more (bad), green: less (better)
Avg On Bottom ROP	ft/h	59.2	42.8	60.2	41.4	68.1	Red: slowest, green: fastest
Bit type	N/A	PDC	PDC	PDC	PDC	Hybrid PDC	

Carbon Footprint Calculations

Scope and Method of Calculations

As illustrated in [Figure 11](#), when direct fuel carbon analysis is unavailable, CO₂ emissions from fuel combustion can be estimated using default average fuel compositions. While the [API Compendium \(2021\)](#) assumes complete combustion for CO₂ estimates, other protocols may apply a fractional carbon conversion. When only facility fuel consumption rates are known, default emission factors can be applied to estimate emissions.

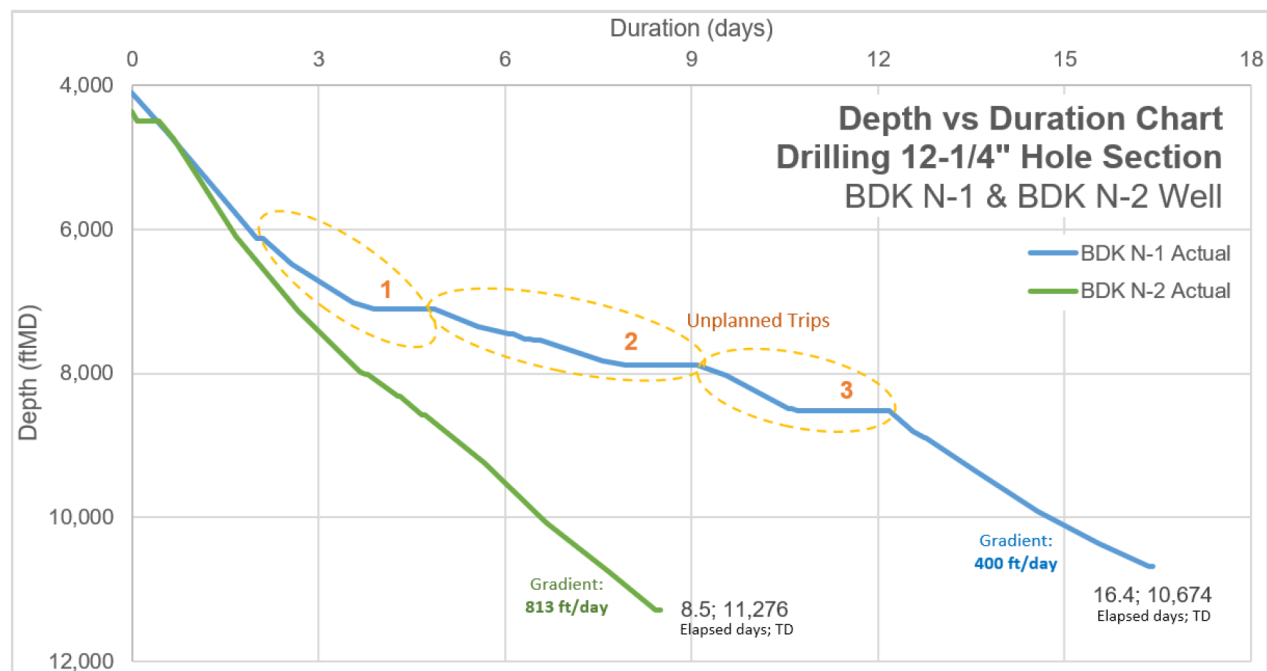


Figure 11—Depth vs. duration chart comparing BDK N-1 and BDK N-2 wells, showing doubled drilling gradient in N-2 after eliminating unplanned trips in N-1.

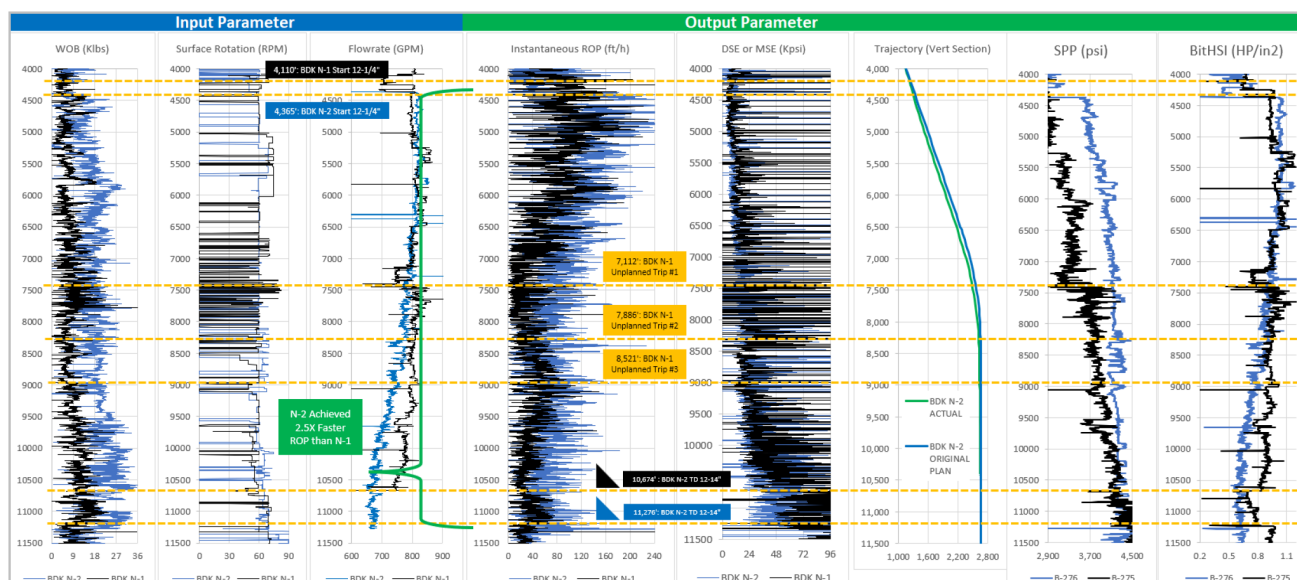


Figure 12—Depth-based drilling parameter plot showing BDK N-2 achieving 2.5 times higher ROP than BDK N-1, drilling 602 ft deeper with lower DSE, confirming improved operational efficiency.

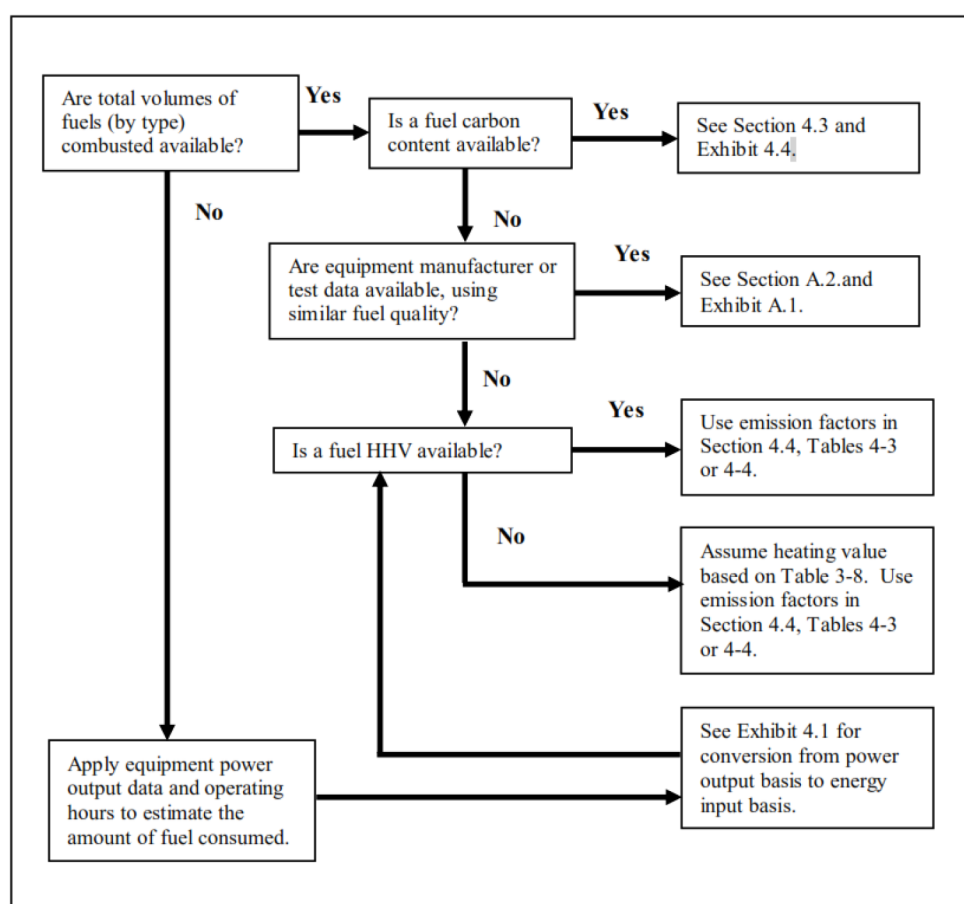


Figure 13—Framework for calculating CO₂ emissions from stationary combustion sources (excluding waste gas disposal) based on API Compendium (2021).

For this study, the emission factor method is applied using Table 4–4 "CO₂ Combustion Emission Factors (Fuel Basis) for Specialized Fuel Types" from the API Compendium (2021), as drilling operations used B30 fuel (30% biodiesel, 70% diesel). In alignment with the Pertamina Standard (Pertamina et al., 2025),

emissions from internal and external combustion units are calculated by multiplying the volume of fuel consumed by an emission factor:

$$E_i = FC \times EF$$

Where E_i : emission of I component (ton); i : calculated emission parameter (CO_2 , CH_4 , N_2O , SO_x , NO_x , PM); FC: fuel consumption (scf or liter); EF: emission factor reference.

Daily fuel consumption is measured via flow meters at the central fuel depot, representing total fuel used for rig power generation and general services (transport vehicles, heavy equipment). Under Pertamina's Tier 3B standard, actual fuel consumption for each combustion unit is recorded, and emissions are calculated directly. Assuming identical equipment specifications, fuel consumption, and thus emissions, is directly proportional to operating duration.

EMISSION CALCULATION REPORT PT PERTAMINA



Sub - Holding : Upstream
 AP/UA/UO :
 Location : Drilling
 Period : September / 2024

Parameter							Emission Factor												Emission, ton								
Code	Well Name	Location	Tier	Emission Factor Reference	HHV, Joule/ton	LHV, Joule/ton	CO ₂	Reference	CH ₄	Reference	N ₂ O	Reference	SO _x	Reference	NO _x	Reference	PM	Reference	CO ₂	CH ₄	CH ₄ (CO ₂ eq.)	N ₂ O	CO ₂ (CO ₂ eq.)	SO _x	NO _x	PM	
Drilling	(B30 Fuel)	Drilling	TIER 3B	Fuel based	4.39E+10	3.76E+10	52.36	Table 4-4 (API Compendium 2021)	0.0021	Table 4-6 (API Compendium 2021)	0.0004	Table 4-6 (API Compendium 2021)	0.2900	US-EPA AP-42	4.4100	US-EPA AP-42	0.3100	US-EPA AP-42	263.389	0.011	0.295	0.002	0.559	0.731	11.122	0.78	
Drilling	(B30 Fuel)	Drilling	TIER 3B	Fuel based	4.39E+10	3.76E+10	52.36	Table 4-4 (API Compendium 2021)	0.0021	Table 4-6 (API Compendium 2021)	0.0004	Table 4-6 (API Compendium 2021)	0.2900	US-EPA AP-42	4.4100	US-EPA AP-42	0.3100	US-EPA AP-42	541.717	0.022	0.606	0.004	1.150	1.504	22.875	1.60	
Drilling	(B30 Fuel)	Drilling	TIER 3B	Fuel based	4.39E+10	3.76E+10	52.36	Table 4-4 (API Compendium 2021)	0.0021	Table 4-6 (API Compendium 2021)	0.0004	Table 4-6 (API Compendium 2021)	0.2900	US-EPA AP-42	4.4100	US-EPA AP-42	0.3100	US-EPA AP-42	554.198	0.022	0.620	0.004	1.176	1.539	23.402	1.64	
TOTAL																				1,359.305	0.054	1.521	0.011	2.885	3.775	57.400	4.035

Figure 14—Example of internal emissions calculation report compliant with Pertamina and international standards, with calculated CO_2 emissions highlighted.

For this paper, the emissions calculation scope covers activities from formation integrity test (FIT) to circulating bottom up (CBU) at section TD. This includes drilling, pipe connections, trajectory surveys, and non-productive duration such as unplanned BHA trips.

Calculation Results

The results in Table 7 demonstrate that bit selection significantly impacted fuel usage within an identical operational scope. In Badak N-2, the Hybrid PDC bit — combining shearing and crushing action — delivered faster drilling progress than the conventional shearing-only PDC bit used in Badak N-1. This improvement reduced drilling duration and lowered fuel consumption by 36%.

Table 7—Daily fuel consumption of Badak N-1 and N-2 wells during the 12-1/4" hole section, with CO₂ equivalent conversion.

Badak N-1			Badak N-2		
Date	Day No	Daily Fuel Consumption (liter)	Date	Day No	Daily Fuel Consumption (liter)
05-Dec-24	1	3,185	03-Feb-25	1	3,815
06-Dec-24	2	7,875	04-Feb-25	2	5,230
07-Dec-24	3	10,460	05-Feb-25	3	7,510
08-Dec-24	4	11,300	06-Feb-25	4	12,600
09-Dec-24	5	11,280	07-Feb-25	5	14,480
10-Dec-24	6	10,170	08-Feb-25	6	16,425
11-Dec-24	7	10,505	09-Feb-25	7	15,895
12-Dec-24	8	11,355	10-Feb-25	8	17,510
13-Dec-24	9	11,730	11-Feb-25	9	16,340
14-Dec-24	10	11,305	12-Feb-25	10	15,965
15-Dec-24	11	10,395	Total Fuel Consumption (liter)		125,770
16-Dec-24	12	11,550	Total CO₂ eq (ton)		225.9
17-Dec-24	13	11,710	Quick converter		
18-Dec-24	14	9,540	Fuel Type Emission Factor, EF (ton CO ₂ eq per liter)		
19-Dec-24	15	12,910	B30 Diesel Fuel: 0.0017962		
20-Dec-24	16	12,890			
21-Dec-24	17	14,000			
22-Dec-24	18	13,540			
Total Fuel Consumption (liter)		195,700			
Total CO₂ eq (ton)		351.5			

Table 8—Summary of CO₂ emission reductions for the Badak N-2 well, based on the daily consumption data in Table 7.

Parameter	Amount	Difference
Total fuel consumption (liter)	69,930	36%
Total CO ₂ eq emission (ton)	125.6	36%

Consequently, CO₂ emissions for the 12-1/4" section in Badak N-2 were substantially lower than in Badak N-1, validating the operational and environmental benefits of optimized bit technology. The direct link between drilling efficiency, fuel savings, and emissions reduction highlights the importance of integrating performance-based bit selection into carbon management strategies. This demonstrated correlation between bit selection, drilling performance, and CO₂ reduction emphasizes that emissions control in drilling operations is not solely a matter of post-activity offsetting, but can be inherently designed into the operational plan. By integrating efficiency-focused drilling practices—such as selecting hybrid bit technology capable of maintaining high ROP in complex formations—operators can achieve measurable environmental benefits without compromising well objectives. These findings from the Badak N-2 case not only provide a proven pathway for lowering operational emissions but also form a replicable strategy for similar mature field developments seeking to balance productivity, cost control, and sustainability objectives.

Conclusion

This paper has demonstrated how integrated formation analysis, thorough pre-job planning, and vigilant execution monitoring can collectively reduce carbon emissions while enhancing drilling efficiency and economic performance. The strategy applied in Badak N-2 saved 7.9 rig days—at a daily rate of approximately US \$76,000—resulting in a direct cost saving of US \$608,000, a significant benefit for marginal field developments while maintaining excellent safety performance.

Drilling bit selection proved to be a pivotal factor in delivering a low-emission drilling operation, particularly in wells with historically challenging interbedded formations. The Hybrid PDC bit, leveraging

both crushing and shearing mechanisms, successfully overcame hard stringers and lithology transitions that typically slow drilling progress—achieving a rate of penetration 2.5 times faster than the previous well. While the bit demonstrated excellent durability in a long interval, real-time monitoring of weight on bit (WOB) against mechanical specific energy (MSE) is recommended to further optimize parameters and extend bit life.

From an environmental perspective, the Hybrid PDC application in Badak N-2 reduced CO₂ emissions by 126 metric tons—a 36% reduction compared to Badak N-1 drilled with a conventional PDC bit. Additional opportunities for emissions reduction remain, such as increasing biodiesel content in fuel (e.g., B40 with 40% biodiesel) to further lower the emission factor while maintaining operational performance. These results affirm that performance-based drilling design can serve as a practical, replicable pathway toward meeting both production and sustainability objectives in mature gas field operations.

References

- Wibowo, Ari, and Ade Diar Suhendar. "Enhanced Petrophysics Analysis to Identify By-passed Zone in Gas Field - A Badak Field Success Story." Paper presented at the SPE Asia Pacific Oil and Gas Conference and Exhibition, Perth, Australia, October 2012. doi: <https://doi.org/10.2118/160305-MS>
- Hinden, P-M., Farrouk, H., Innes, G., Crowther, L., and S. Kustanto. "Reducing Carbon Emissions in Drilling Operations Offshore Abu Dhabi." Paper presented at the ADIPEC, Abu Dhabi, UAE, October 2023. doi: <https://doi.org/10.2118/216172-MS>
- Khan, Shahjahan, Faisal, Muhammad, and Asjad Javed. "Drilling Towards a Greener Future: Enhancing Efficiency and Reducing Carbon Footprint." Paper presented at the SPE/PAPG Pakistan Section Annual Technical Symposium and Exhibition, Islamabad, Pakistan, February 2025. doi: <https://doi.org/10.2118/226836-MS>
- Sayogyo, Bramarandhito, Hermawan, Aditia, and Bastian Andoni. "Delivering a One Phase Well Under Two Days with a One Inch Cutter PDC Drill Bit: A Record Breaking Performance in Mature Field, Drilling Optimization Case History." Paper presented at the Offshore Technology Conference Asia, Virtual and Kuala Lumpur, Malaysia, March 2022. doi: <https://doi.org/10.4043/31638-MS>
- Moi, Sonja, Carlsen, Liv Almas, Skadsem, Hans Joakim, Islam, Aminul, and Ole Anders Helstrup. "Nonlinear Regression Analysis and System Stiffness Approach for Formation Integrity Test Interpretation." Paper presented at the SPE/IADC Drilling Conference and Exhibition, The Hague, The Netherlands, March 2017. doi: <https://doi.org/10.2118/184602-MS>
- Rushmore, Peter "Anatomy of the "Best In Class Well": How Operators Have Organised the Benchmarking of their Well Construction and Abandonment Performance." Paper presented at the SPE/IADC Drilling Conference and Exhibition, Amsterdam, The Netherlands, March 2011. doi: <https://doi.org/10.2118/140172-MS>
- Armenta, Miguel. "Identifying Inefficient Drilling Conditions Using Drilling-Specific Energy." Paper presented at the SPE Annual Technical Conference and Exhibition, Denver, Colorado, USA, September 2008. doi: <https://doi.org/10.2118/116667-MS>.
- Mayibeki, D., Hareland, G., Al Dushaishi, M., and S. Blaylock. "The Effect of Confinement on Rate of Penetration (ROP) in Hard Rock." Paper presented at the 57th U.S. Rock Mechanics/Geomechanics Symposium, Atlanta, Georgia, USA, June 2023. doi: <https://doi.org/10.56952/ARMA-2023-0813>
- American Petroleum Institute. (2021). Compendium of Greenhouse Gas Emissions Methodologies for the Natural Gas and Oil Industry (API Compendium 2021). American Petroleum Institute.
- Pertamina Standard. "Pertamina Standard: Emission Calculation Standard". Jakarta, Indonesia, 2025.